

UGUT, Russian Federation

WMO: 239460

Lat: 60.500N

Lon: 74.017E

Elev: 158

StdP: 14.61

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.3	-33.2	-44.0	0.4	-38.1	-38.3	0.6	-32.5	13.9	14.9	12.7	13.2	2.1	240	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	84.7	65.7	81.3	64.8	77.8	63.4	69.3	78.3	67.4	76.4	65.8	73.8	6.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	97.6	71.9	64.5	91.4	70.7	62.6	85.5	69.7	33.6	78.5	32.0	76.7	30.8	73.5	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	12.7	11.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-45.4	72.3	2.8	3.1	-47.5	74.6	-49.1	76.4	-50.7	78.1	-52.8	80.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.7	-4.2	1.2	17.7	32.9	44.0	60.9	64.7	58.4	46.2	31.0	10.5	2.8
	DBStd	26.51	16.65	16.59	12.63	10.07	10.02	9.44	6.86	7.05	7.67	9.49	15.31	16.43
	HDD50	8248	1681	1366	1000	516	247	21	0	13	164	592	1184	1464
	HDD65	12697	2146	1786	1465	962	655	186	92	223	565	1054	1634	1929
	CDD50	1195	0	0	0	4	60	348	455	275	50	3	0	0
	CDD65	169	0	0	0	0	4	63	82	19	1	0	0	0
	CDH74	1672	0	0	0	0	77	651	781	161	2	0	0	0
	CDH80	427	0	0	0	0	14	178	213	22	0	0	0	0
	CDH80	427	0	0	0	0	14	178	213	22	0	0	0	0
(14) Wind	WSAvg	5.2	4.4	4.5	5.7	6.1	6.2	5.2	5.2	4.6	5.0	5.4	5.2	4.5
(15) Precipitation	PrecAvg	23.7	1.2	1.0	1.3	1.3	2.2	2.8	2.6	3.2	2.5	2.0	1.9	1.5
	PrecMax	31.9	2.6	1.8	3.7	2.6	6.4	6.3	5.4	6.7	4.7	4.2	2.9	3.6
	PrecMin	18.0	0.2	0.4	0.0	0.0	0.8	0.1	0.3	0.8	0.5	0.7	0.6	0.5
	PrecStd	3.9	0.6	0.4	0.7	0.6	1.2	1.1	1.3	1.5	1.2	0.9	0.6	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	36.1	43.3	63.8	81.9	88.7	90.2	82.9	72.4	60.5	37.6	32.0
		MCWB	29.4	33.0	34.1	52.9	60.0	67.3	70.4	66.1	57.9	49.0	36.5	31.0
	2%	DB	26.1	32.1	38.7	56.0	74.4	84.8	85.2	78.0	66.3	51.5	34.7	29.9
		MCWB	25.1	30.7	32.7	46.7	58.0	67.0	64.8	64.0	55.4	46.4	33.7	28.9
	5%	DB	22.2	28.1	35.9	51.4	68.5	81.4	82.0	75.0	61.6	47.0	32.3	26.9
		MCWB	21.2	26.6	31.7	43.1	55.8	65.7	64.6	63.1	53.6	44.4	31.5	25.9
	10%	DB	18.6	23.1	33.8	47.8	61.0	77.5	78.4	71.9	57.7	43.1	30.6	23.1
		MCWB	17.7	21.6	30.4	40.6	51.8	63.9	63.7	62.2	52.4	40.5	29.5	22.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.2	33.4	36.2	53.2	63.4	73.5	72.3	69.1	60.1	50.1	36.8	31.4
		MCDB	29.5	35.1	40.4	62.0	78.3	82.1	86.2	76.8	67.0	56.2	37.3	31.8
	2%	WB	25.0	30.7	34.1	48.2	59.4	69.8	69.2	67.0	57.2	46.8	33.8	28.9
		MCDB	26.1	32.1	37.6	54.5	70.4	79.2	77.7	74.1	62.8	50.6	34.6	29.8
	5%	WB	21.2	27.0	32.3	44.2	56.1	67.8	67.3	65.3	55.1	44.2	31.7	26.0
		MCDB	22.1	28.3	35.5	50.7	67.3	78.7	75.4	72.2	60.3	47.5	32.3	26.8
	10%	WB	17.7	21.6	30.4	41.0	52.7	65.8	65.8	63.1	52.7	40.8	29.6	22.4
		MCDB	18.5	23.1	33.5	46.3	60.9	74.6	73.5	69.5	57.7	42.7	30.5	23.3
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	16.4	17.3	17.1	17.5	18.4	18.3	16.9	13.1	9.3	13.1	12.2
		MCDBR	14.0	14.8	14.9	22.1	28.2	24.5	25.0	22.7	18.8	13.4	10.9	11.7
		MCWBR	13.7	14.2	11.3	14.2	14.7	10.6	8.6	10.7	9.8	9.6	10.9	11.4
		MCDBR	14.1	14.6	13.3	21.2	26.3	22.0	19.4	18.7	16.7	13.0	11.6	10.8
	5% WB	MCWBR	13.7	14.1	10.8	14.6	15.7	11.3	8.2	10.6	10.1	9.8	11.5	10.7
(40) Clear-Sky Solar Irradiance	taub		0.264	0.306	0.359	0.379	0.392	0.403	0.425	0.413	0.362	0.331	0.278	0.250
	taud		1.979	2.008	1.947	2.065	2.277	2.327	2.259	2.266	2.370	2.261	2.052	1.923
	Ebn at Noon		141	205	231	255	263	262	252	244	238	200	135	93
	Edh at Noon		15	27	40	43	37	36	38	35	26	21	14	10
(44) All-Sky Solar Radiation	RadAvg		139	426	891	1373	1522	1680	1603	1138	709	311	134	71
	RadStd		12	36	66	132	134	157	178	110	83	36	28	7

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-662	N/A	N/A	

Nomenclature: See separate page